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Artificial Intelligence for Modelling Infectious Disease Epidemics

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Abstract

Infectious disease threats to individual and public health are numerous, varied, and frequently unexpected. Artificial intelligence (AI) and related technologies, which are already supporting human decision making in economics, medicine and social science, have the potential to transform the scope and power of infectious disease epidemiology. In this review, we consider the application to infectious disease modelling of AI systems that combine machine learning, computational statistics, information retrieval, and data science. We first outline how recent advances in AI can accelerate breakthroughs in answering key epidemiological questions and we discuss specific AI methods that can be applied to routinely-collected infectious disease surveillance data. Second, we elaborate on the social context of AI for infectious disease epidemiology, including issues such as explainability, safety, accountability, and ethics. Finally, we summarise some limitations of AI applications in this field, and provide recommendations for how infectious disease epidemiology can harness most effectively current and future developments in AI.

Main

Artificial intelligence [1] (AI) is transforming many aspects of contemporary science [2] and has the potential to similarly change the landscape of infectious disease epidemiology. AI can be defined as intelligent behaviour exhibited by machines and computers and has been an active area of research since the 1950s [3]. Over the past decade the focus of AI methods has shifted substantially, from logic-based approaches [4] to those associated with deep learning [5]. In this perspective, we define AI and related data science approaches broadly and therefore include methods from machine learning (ML) [6], probability theory [7], numerical optimisation [8] and new directions in scalable computation [9,10].

Infectious disease epidemiology is the study of why infectious diseases emerge, how they transmit within and among populations, and of the strategies that can be used to prevent, control, and mitigate their spread [11]. Mathematical, computational and statistical modelling is an essential component of this interdisciplinary field, and quantitative models are used to inform public health policies and responses at local and global scales [11]. Although much attention has been paid to the application of AI to problems in human health, such as patient diagnosis [12], individual-level disease risk prediction [13] and decision-support for doctors [14], there have been fewer demonstrations of the utility of AI models for infectious disease epidemiology – mirroring the lower uptake of AI methods in population health more generally, including for non-communicable diseases [15]. Yet, effective and sustainable improvements to public health must address determinants of health at both the individual and population levels, as well as the interactions between them [16]. New AI approaches that span public health and individual outcomes have the potential to transform infectious disease responses and enable more targeted, equitable and robust interventions.

One possible reason for the relatively lower penetrance of AI in infectious disease epidemiology is the challenge of acquiring large-scale, standardised and representative data [17] for the training and evaluation of AI/ML models with many parameters. However, new AI approaches perform increasingly well with limited data. For instance, approaches based on **fine-tuning** or **transfer learning** [18] (Box 1) of large pre-trained networks can achieve strong performance without the need for months of initial training or terabytes of data, and recent advances in **self-supervised learning** [19] can facilitate further **zero-shot learning** on a range of epidemiological questions (Box 2).

Box 1: Glossary.

Active learning (AL) is a machine learning framework designed to minimise data labelling efforts; the model iteratively asks a human user to label relevant samples and the model learns from the replies.

Bayesian optimization is used to numerically optimise a function that is difficult to optimise or evaluate using conventional approaches.

Causal inference refers to a diverse suite of statistical methods that investigate and exploit the cause-effect relationships that underlie observed associations.

Explainable AI is a set of methods that allows human users to comprehend the output created by ‘black-box’ machine learning algorithms (i.e. algorithms and models whose internal workings are not easily accessible or interpretable).

Deep generative models are machine learning models that use deep neural networks to generate/simulate new data that are similar to the training data.

Fine-tuning refers to the training of a previously fitted model (i.e. a **pre-trained** model) on a new task. It is a type of **transfer learning** and is typically employed where data are scarce.

Foundation models refers to very large deep learning models that are pre-trained on vast, general-purpose, data sets. Examples include **large language models (LLMs)**.

Graph Neural Networks (GNNs) are a family of artificial neural networks for processing graph-structured data (e.g. social networks or protein structures).

Mechanistic epidemic models are models in which a plausible mechanism for the infection process is represented in an analytical or simulation framework (e.g. using differential or difference equations).

Multimodal learning refers to approaches that attempt to leverage multiple types of data, such as images and language.

Recurrent neural networks (RNNs) are a family of artificial neural networks designed for processing temporal, sequential data. They have been used extensively for problems in speech, natural language, and genetics.

Reinforcement learning (RL) is a machine learning paradigm in which an agent learns how to take actions conditioned on its state in an environment, so as to maximise a reward signal. With human feedback, these systems can be aligned to match human preferences.

Retrieval-Augmented Generation (RAG) is a technique to assist l**arge language models** for text generation.

Self-supervised learning is a ML method in which a model learns from data without labels for guidance (c.f. **active learning**).

Surrogate model (or emulation) is a model used to approximate another, typically more complex or computationally expensive, model.

Transfer learning is a process whereby the knowledge acquired by a model from an initial task is leveraged for a later task.

Transformers are neural network architectures that process sequences (e.g. text) in parallel and compute the relationships of each part of the sequence to others, enabling an understanding of the context of that sequence.

Zero-shot learning refers to the ability of an AI/ML model to generalise to a new task or problem that is not represented in the data set used to train the model.

Box 2: Important Epidemiological Concepts.

Fatality Ratio: The proportion of those affected by a disease who do not survive, calculated for different subpopulations (e.g. for cases/reported infections or hospitalizations) and often stratified by factors such as age, time, and location. The infection fatality ratio (i.e. the proportion of those infected who subsequently die from that infection) is an important population-wide measure of disease severity. Given the time lags inherent in both the infection process and health reporting systems, it is not appropriate in an early, growing epidemic to estimate the infection fatality ratio as the number of deaths to date divided by the number of infections to date and parametric modelling is required [161].

Forecasting: The projection of a disease's future trajectory using mathematical and/or statistical tools. Generally, trends in summary statistics such as incidence or prevalence are projected forward in time. Many approaches blend a mix of time series modelling with mechanistic models of disease spread.

Generation Interval: The duration between the infection of an individual and the infection of one of its secondary infections; this length of time varies among infector-infectee pairs and hence is represented by a probability distribution. This value depends on factors such as the duration and timing of infectiousness and may be affected by the epidemic growth rate or local prevalence of infection [162]. Parameters such as the generation interval and the **reproduction number** strongly determine the growth rate of an epidemic and are used to determine the planning and timing of public health interventions. Precise infection times are almost never directly observed, hence the generation interval distribution is usually estimated by a combination of modelling and partial observation.

Infection Networks: The spread of a contagion through a population, represented as a weighted, directed network between infectors and infectees [163]. While true, complete infection networks are rarely observed directly, they can be estimated through a mixture of epidemiological and contact pattern data. Key epidemiological parameters can then be estimated using generation intervals and other data, such as pathogen genome sequences or demographic information.

Nowcast: The reconstruction of the current state of the epidemic as accurately as possible, when contemporaneous data may be affected by reporting delays. Even in the absence of delays, integrating information from heterogeneous sources requires data synthesis. Nowcasting uses statistical models to debias and synthesise multiple, noisy data sources to present a current picture of an epidemic.

Reproduction Number: The average number of secondary cases arising from a single case in a population, also known as the R value. If estimated for a situation where all individuals are susceptible to infection, this is termed the basic reproduction number, R_0. The effective, or time-varying reproduction number, R_t, relaxes this assumption and considers the average number of cases generated in the current state of the population. The reproduction number is typically estimated from data using methods including regression, mechanistic models or networks [164].

Transmissibility: The propensity of a pathogen to be transmitted between individuals, which is a combination of pathogen-related factors relating to the pathogen (e.g. the size of the inoculum and the pathogen's ability to evade host immune responses) and host-related factors (e.g. the number and intensity of contacts a host makes). Capturing change in transmissibility over time is important for planning interventions.

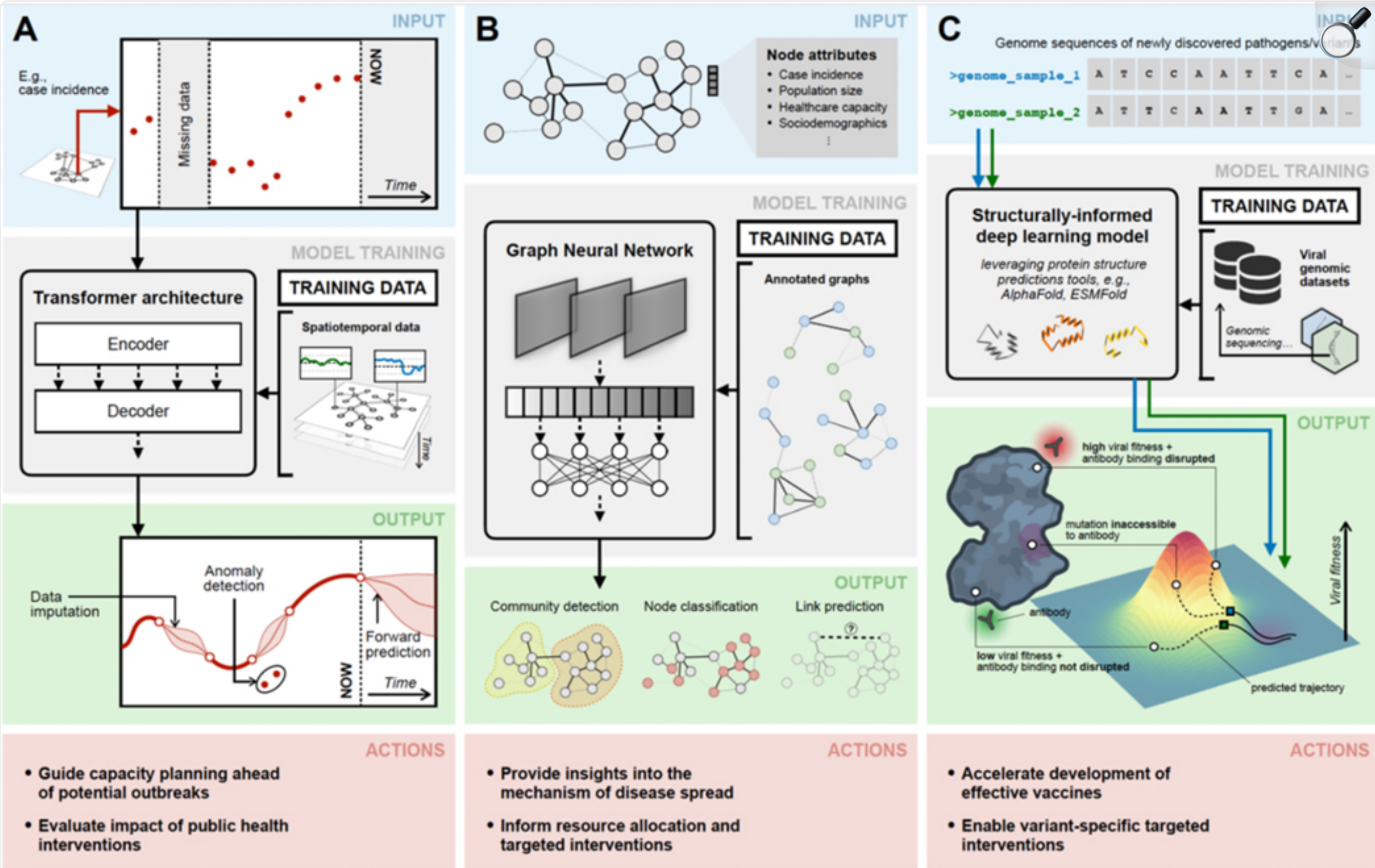
Vaccine effectiveness is a measure of how well vaccination protects against infection, transmission, and/or symptomatic and severe disease, typically measured as one minus the relative risk of the outcome in vaccinated vs. unvaccinated persons. When measured in a randomised trial, the term "vaccine efficacy" is often used.

We consider two main topics in this perspective: first, we outline how current and anticipated advances in AI can change how we answer key questions in infectious disease modelling. For example, novel AI methods can improve the collection and integration of heterogeneous data sources, and AI/ML could be included in policy and decision-making frameworks to improve population health. Second, we discuss some social and ethical contexts of the application of AI to infectious disease epidemiology; we consider the explainability, safety, and accountability of AI in public health, and provide recommendations for how infectious disease epidemiology can harness developments in AI effectively.

AI to answer epidemiological questions

One of the key challenges during the early stages of an epidemic is to understand the severity and epidemic potential of the infectious agent. This involves estimating fatality ratios, serial and generation intervals, transmissibility, and epidemic growth rates, as well as inferring infection networks and transmission heterogeneity in different settings (Box 2) [20]. In traditional epidemiology, these questions are often answered using observational data (e.g. case-control studies, cohort studies or household surveys) [21]. However, the small scale of such studies and the idiosyncrasies of data collection mean that the true transmission process is observed imperfectly and may not be representative of the whole epidemic. The actual chain of infection events and locations where infections occur is often uncertain (e.g., individuals may visit multiple locations and contact many people, some of whom might be infectious but asymptomatic), hindering efforts to accurately measure quantities such as the incubation period or transmission intensity from observational data alone. Other issues include underreporting, censoring, truncation, non-random data omission, and uneven data reporting. Here, Bayesian data augmentation [22] has proven valuable in improving parameter inference in the context of missing data, and AI approaches can help the scalability and inference of such models [23] (Fig. 1A). A promising direction involves approximate Bayesian inference such as that based on variational inference with normalising flows [24]. This reframes the problem from one of sampling to optimization, which can be addressed efficiently using gradient-based deep learning techniques. Use of these tools makes fast and accurate inference more achievable, especially when inferences have to be completed quickly.

Figure 1: AI approaches to tackling key epidemiological questions.



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(A) Time-series analysis using foundation models (FMs). Pre-training large-scale FMs with transformer architecture using observations (empirical or simulated) from time-series epidemic data allows zero-shot applications ([Box 1](#)) for forward prediction, data imputation, and anomaly detection. Output from these models can be used to evaluate the impact of public interventions and guide capacity planning for future outbreaks. **(B)** Modelling infectious disease spread using Graph Neural Networks (GNNs). Pathogen dissemination can be represented via annotated graphs, where nodes correspond to locations or individuals and edges represent potential transmission pathways (e.g., human or vector interactions); each node is associated with a set of features (e.g., case incidence, population size) that are either indicators or drivers of spread. GNNs are capable of learning complex patterns from such data, enabling node classification (predicting disease prevalence), community detection (identifying infection clusters), and link prediction (revealing cryptic transmission pathways). These insights can provide a detailed understanding of the underlying mechanism of disease spread and inform resource allocation and targeted interventions. **(C)** Predicting immune escape mutations using biologically-informed deep learning models. By leveraging recent advances in protein structure prediction (e.g. AlphaFold, ESMFold), these models could enable the early detection of pathogens or variants likely to develop mutations conferring resistance to existing vaccines or therapeutics through antibody-binding disruption. Such predictions can inform the design of next-generation vaccines and guide the prioritisation of containment strategies targeted at emerging lineages.

Given the difficulties in generating a complete description of the transmission tree that underlies an epidemic, model-based analyses are performed on routinely collected aggregated data, such as counts of cases, hospitalisations, or deaths. Mechanistic and semi-mechanistic disease transmission models are commonly used to test hypotheses and estimate key epidemiological parameters (e.g., transmissibility, virulence) with associated uncertainty [25]. Whilst these models provide substantial mechanistic insights into transmission and can be utilised to construct counterfactual scenarios, they often come with significant computational costs due to the complexity of the numerical methods and inference in a high-dimensional parameter spaces [26]). Recent advances in deep generative modelling (Box 1) have shown that inference can be accelerated by approximating the original model with a generative model surrogate [27,28] or by using variational inference [29] (Box 1, Table 1). With improvements in the speed of model computation comes additional opportunities to increase the model complexity and realism, including potential linkages between individual transmission heterogeneity and population-level outcomes [30]. Because AI-accelerated methods have the potential to reduce model run times, from weeks to hours, they could significantly speed up the iterative process of informing policy decisions and model refinement (Table 1).

Table 1:

New AI approaches to answering key questions in infectious disease epidemiology, their potential impact, and level of maturity in the scientific and public health community.

Problem domain	Method	Potential impact	Level of maturity
Inference of epidemiological parameters	Generative Bayesian and surrogate models [36]	Incremental improvements in speed and accuracy of epidemiological estimates, especially when data are missing. Faster generation and iteration of model results for policy.	Bayesian methods can be implemented easily in available software packages. Surrogate models need to be developed for specific applications[153]
Epidemic forecasting/nowcasting	Time series foundation models [154] and ensemble techniques [39]	Potential for improved speed and accuracy in estimating future trajectory of cases. Better generalisation of trends for medium term forecasts.	Ensemble forecasting is standard practice in influenza and COVID-19. Several foundation time series models exist and are easy to access.
Scenario modelling	Compartmental, mechanistic models [155] and reinforcement learning agent-based models [156]	AI's impact may be limited because mechanistic models better capture the impact of interventions. AI models can be used for reinforcement learning [156] for de novo discovery of individual	Reinforcement learning neural network architectures exist, but bespoke versions need to be created for specific applications.

Problem domain	Method	Potential impact	Level of maturity
		behavioural adjustments and policy options, and to speed up the generation of realistic scenarios.	
Understanding mechanisms of disease spread	Graph Neural Networks [94] and Graph Foundation Models [60]	Ability to incorporate large amounts of structured data with the potential for improved inference of disease transmission routes.	Early research outputs
Infectious disease surveillance	Active Learning [157] and Bayesian Optimization	Improved cost-effectiveness	Early research with limited implementation
Risk prediction	Multimodal AI [158]	Improved inference of risk factors of severe disease from multiple data types	Early research with potential applications in population health
Analysing pathogen genomes	Protein Language Models (PLMs) [71,159]	New applications are possible for improved design of vaccines and therapeutics, and for anticipating and evaluating pathogen variants with new phenotypes (e.g. immune evasion).	Early research outputs
Public health decision making	Markov Decision Processes and Reinforcement Learning [160]	Improved ability to *learn* optimal decisions from human-machine interactions	Conceptual, theoretical

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Prediction & forecasting

During an infectious disease epidemic, policy decisions regarding the deployment and timing of interventions rely on reliable estimates of the number of current cases and forecasts of future cases [31]. For a given disease, the number of cases, hospitalizations, and deaths reported by public health surveillance systems over time are analysed to inform key policy decisions. While epidemiological data sets may include a rich variety of demographic (e.g., ethnicity, age), clinical, geospatial, and genomic information, at their simplest they consist solely of quantitative trends and can be therefore analysed using a range of approaches, including statistical time-series techniques [32], modern ML methods and advanced deep learning models [5] including newer **foundation models** (Table 1).

However, surveillance data cannot tell us directly how many individuals are infectious at any given day, or what the expected future trajectory of cases might be. Epidemic surveillance data are almost always noisy and affected by biases in reporting (e.g. time-variable delays), testing (e.g. targeting of limited capacity or willingness to test in certain groups), and sampling (e.g. towards groups that can more readily access healthcare) [33,34]. For example, data may appear to suggest a declining epidemic, but this trend could be an artefact of time-dependent changes in reporting [35]. These data issues could lead to ineffective and delayed responses.

Current approaches to **nowcasting** [35–37] attempt to address these systematic biases and enable robust estimation of the contemporary epidemic situation [38]. However, choosing an optimal nowcasting model, which can both generalise to new data and accurately capture complex patterns, is challenging, especially from noisy and incomplete data. The simplest solution is to choose a model from a large, complex class of models and then try to mitigate any potential overfitting or over-parameterisation (Fig 1A). Recent work in which nowcasting and forecasting are improved with ensemble methods shows promising results and provides successful avenues to create model diversity [39,40]. During the COVID-19 pandemic multiple research groups undertook and evaluated epidemic forecasts, accelerating progress towards more standardised and rigorous models for decision making.

Foundation models from large deep neural networks [41] represent new and powerful ways to explore and understand time series surveillance data and go beyond current approaches. However these foundation models need to be pretrained on hundreds of billions of data points from a diverse range of time series (across a range of domains); consequently the patterns they learn are effective at zero-shot [42] (Box 1) performance on a range of new surveillance data. The models restrict the fitted functions to plausible shapes, helping to accurately quantify uncertainty [42]. Historically, AI models were unable to include rigorous measures of uncertainty but recent work in deep learning shows promise in using activation functions [43], ensembles [44] and conformal prediction [45] for the quantification of uncertainty.

Modelling mechanisms of disease spread

Infectious disease modelling has relied traditionally on mechanistic models which represent the disease transmission process as a simplified but definable process. Examples include mathematical Susceptible-Infected-Recovered (SIR) equations, and individual/agent-based models [46], in which a simulation of transmission is constructed and each agent is assigned specific attributes and events. When fitted correctly to rich data individual based models generate insights into the transmission process and can be used to evaluate targeted interventions. However, parameter inference is much harder to perform using agent-based models than for standard statistical techniques due to their lack of analytical components [29]. Here, AI approaches are providing new ways to facilitate efficient inference [29] through a range of innovative approximations that employ deep neural networks. For example, by leveraging variational Bayesian inference new neural network architectures can accommodate large and dynamic agent-based models [47]. Without AI approaches, inference on very large agent-based models is challenging as the parameter spaces are too large, and generating samples or realisations from such models can be time consuming an alternate solution to computational intractability is provided by gradient-based methods such as variational inference, as discussed above.

Dynamical 'compartment' models are one of the most commonly-used analytical tools in mathematical epidemiology, the best known of which is the SIR model, which partitions the population into a small number of groups (compartments). SIR dynamical models are comparatively easy to fit to data but often lack the detail necessary to bridge the gap between individual behaviour and population-level dynamics. Alternative models based on branching processes [48], partial differential equations, or self-excitatory Hawkes processes [49] are growing in popularity and there are strong similarities between these approaches [48,50]. As previously noted, foundation time series models can be used to constrain function space by restricting the set of functions *a priori* and embedding these constraints within a chosen mechanistic model (termed semi-mechanistic models). The overall objective is to base modelling on plausible epidemiological realities. However model misspecification can still introduce errors. Using these approaches AI models have the potential to not only achieve high predictive accuracy within current mechanistic frameworks, but also to learn more complex hidden mechanisms that can improve existing models.

Graph Neural Networks (GNNs) that operate on discrete-structured data (Fig. 1B) [51] present a particularly promising type of AI for detailed forecasting of infectious disease dynamics [52]. Graphs (or networks) emerge naturally in many areas of infectious disease epidemiology, including contact networks of the spread of a disease through individuals and populations [53], phylogenetic trees and networks to track pathogen evolution, and social and information networks to understand health behaviours (e.g., masking, vaccine uptake), the spread of information and misinformation, and social influence [54]. Through their ability to represent rich relationships, GNNs have been shown recently to accurately predict COVID-19 cases per region [55], forecast influenza-like illness [56], and nowcast vaccine uptake from online information networks [57]. GNNs also show promise in expanding dynamical compartmental models to capture complex spatio-temporal dynamics among discrete geographical regions [58]. GNNs can expand agent-based models by allowing differentiability of all model components that facilitates joint inference over complex models [59]. Finally, future graph-based foundation models may be able to learn transferrable representations of graphs that generalise to any unseen graph [60], thus potentially enabling the transfer of knowledge from data rich to data poor settings. Many important discrete problems on graphs are NP-hard (nondeterministic polynomial time - finding a solution is impossible given current knowledge) and need to be solved heuristically. AI approaches via GNNs offer powerful continuous optimisation tools such as gradient descent, which can improve the solving of discrete problems on graphs.

Immunological and genomic interactions

Disease preparedness and response requires the timely detection of emerging zoonotic pathogens or novel variants of known pathogens. Even though genomic surveillance is essential for monitoring the circulating genetic diversity, determining the epidemiological and disease phenotype of the pathogens identified usually requires time- and resource-consuming experimental work. Advances in AI models change this paradigm by utilising genetic sequences as their input, capturing information about pathogen proteins [61,62]. These can pave the way for high-throughput prediction of protein structures and pathogen phenotypes from genomic data, enabling rapid preliminary assessment of new phenotypes and ultimately reducing the amount of experimental work needed to obtain good predictive performance [63] (Fig. 1C). However, predicting from small labelled data sets to large unlabelled data sets can be error-prone and recent cross-prediction approaches show great promise in improving downstream inferences [64]. AI models applied to genomic data can be also used to classify virus lineages[65], infer when, where, and how a pathogen emerged [66], predict the pathogen traits such as transmissibility [67], escapability and spread [68], and predict host specificity in order to identify likely cross-species spillovers [69,70]. AI models can also enhance the accuracy of phylogenetic inference, enabling a more precise characterisation of the infection process.

In addition to predicting complex and multifaceted phenotypes from genetic sequences, AI approaches can help infer the evolutionary trajectory of pathogens already circulating in a host population and can potentially anticipate escape variants [71]. For example, human respiratory viruses that cause great global health burden, such as SARS-CoV-2 and influenza viruses, accumulate mutations continuously due to replication errors that, through natural selection, allow them to evade host antibody immunity mounted against previous virus variants [72] (Fig. 1C). AI-driven variant forecasting can inform the development of targeted vaccines, monoclonal antibodies and drugs targeting future pathogen strains [73,74] could be paired with recent advances in accurately predicting protein structures as monomers and in complex with other molecules [75]. However, the variant that will dominate global or local spread will not only depend on predictions of pathogen evolution, but also on underlying dynamical processes influenced by human behaviour, cross-immunity, climatic conditions, and prior exposure to other, related pathogens [76]. Failing to account for the complexity of these interacting processes could lead to false predictions from AI models alone. For example, recent work showed that epidemiological features predict SARS-CoV-2 variant spread better than those based on evolutionary and immunological drivers [77].

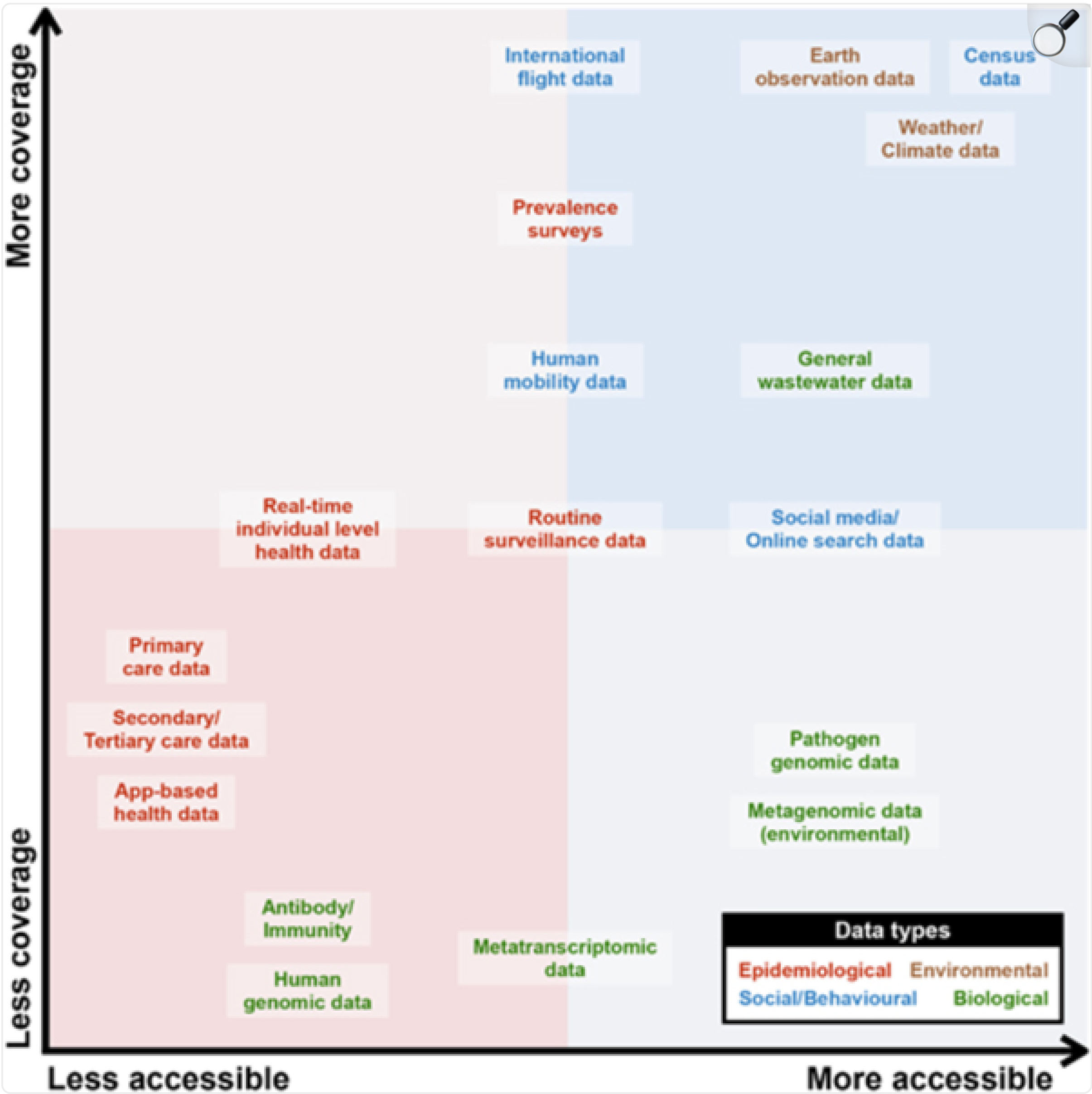
Sequence-based AI methods also enable exploration of as-yet unseen pathogen diversity. The recent metagenomic and metatranscriptomic discovery of novel RNA viruses has exponentially expanded our knowledge of the diversity of viruses found in animals, some of which could potentially spillover into humans and cause disease, or interact with other co-infecting pathogens [78]. The default tool for studying this diversity is molecular phylogenetics, which comprises a diverse field [79] of methods spanning the biological, statistical and computational sciences. Variational Bayesian inference empowered through probabilistic programming could deliver a step change in the scale of phylogenetic tree estimation possible from molecular sequences [80]. Further, ML/AI models may be used to improve predictions of spillover potential between reservoirs and human hosts using pathogen genomic data [78].

AI-aided integration of key data

Data science & representative sampling

Despite a step-change since the COVID-19 pandemic in our ability to perform large-scale, interdisciplinary data collection during disease outbreaks, data scarcity and geographical biases remain major obstacles in global epidemiological research (Fig. 2) [81]. These have been shown to exacerbate racial biases [82] and lead to non-optimal public health policy decisions. In particular, the uneven and inequitable distribution among locations of diagnostic testing resources (including genotyping) [83] results in biased sampling, limiting the representativeness and utility of the data collected [84]. It is not only the volume of data that necessarily provides the most insights, but the quality of it (e.g., sampling design, representativeness); this Big Data Paradox [85] has important implications for our ability to infer key epidemiological parameters, to make accurate forecasts, and to respond to emerging threats [86].

Figure 2: Classification of data types to investigate infectious diseases.



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Data used to inform epidemiological modelling are placed in terms of their accessibility to the research community and their population level coverage. Data are classified based on their data types, including epidemiological (red), environmental (brown), social-behavioural (blue), and biological (green).

As the global community continues to scale up infrastructure for infectious disease data collection and integration of data types that help understand the drivers of transmission (e.g. socio-economic, census, satellite derived climate/environmental data, Fig. 2), AI might be used to tackling problems such as improving the cost-effectiveness of surveillance and mitigating uneven representation in surveillance data. For example, **Active Learning**[87] and **Bayesian Optimization**[88] (Box 1, Table 1) can be used to inform the design of adaptive disease surveillance strategies (e.g., selecting the next set of locations or individuals to test based on current observations) that are tailored to answer specific epidemiological questions [89]. They are particularly useful when the costs of data collection and processing are high, as is the case of genomic surveillance, for example. However, implementing an adaptive surveillance system for outbreak detection, tracking and monitoring is challenged by the logistics and time-scales for data collection, processing, and translation into actionable insights [89]. Tightly linking computational and practical approaches to disease surveillance will be necessary if AI models are to offer tangible benefits. With openly accessible data sets on the drivers of transmission it might become possible using AI to predict disease distributions without access to highly resolved epidemiological data.

Integrating novel data sources

A plethora of new data sources and digital tools is creating new approaches to the modelling of infectious disease epidemics. During the COVID-19 pandemic in the UK, the National Health Service Covid-19 App was downloaded to >21 million mobile phone devices, providing users with timely notifications of infectious disease exposures and instructions in case their contact was diagnosed positive [90]. Such mobile applications are a major new source of individual-level behavioural data for understanding and informing public health interventions. Research has shown how digital contact tracing apps could transform our ability to reduce transmission at a fraction of the cost of manual contact tracing [91], especially when pathogens transmit rapidly and only a minority of infected individuals show early symptoms. Further, contact tracing apps could provide new, real-time data about the variation in transmissibility as a function of proximity and duration of contact, and how this variation changes through time and among pathogen variants [92].

More generally, it is now feasible to aggregate individual-level mobility information from mobile phones, and this new source of data can substantially enhance the precision and utility of mathematical models of infectious disease [93,94]. For example, such data is used to characterise population-level trends in human movement during emergencies [95], and to understand how frequently individual venues are visited by individuals. During the COVID-19 pandemic a variety of sources of smartphone-derived mobility data were used to monitor population behaviours and how they shifted in response to public health interventions or perceived risk. Data governance and privacy-preserving aggregation of location signals can ensure that individuals are not identifiable, but challenges remain in ensuring that sensitive data are not leaked or used unintentionally to reinforce existing biases. Whilst mathematical descriptions of how pathogens spread through networks of susceptible, infected and recovered individuals are well established, further work is needed to integrate these epidemiological principles with new data sources and move beyond describing patterns in big data sets, especially when the contact networks that underlie transmission are transient and highly structured [86]. GNNs (see section above) provide a new approach to the linkage of high-precision epidemiological models with novel data sources.

Integrating individual- and population-level indicators of disease activity using combinations of wastewater surveillance, population surveys, syndromic and digital surveillance, body-worn physiological sensor data (e.g. heart rate, step counts, sleep), and contact network information could enhance our ability to detect outbreaks, monitor disease spread, and follow the long term sequelae of outbreaks [96]. For example, sensory-based surveillance data can detect early signs of long COVID from sleep and physical activity patterns [97]. Reinforcement learning (RL) systems (Box 1) have been used to support testing decisions at airports when limited testing was available [98]. There is the potential to apply multimodal AI to images, text, speech, and other data to interrogate and predict outcomes at the individual and population level [99].

In addition to consensus genome sequences of sampled pathogens, the genetic diversity captured in metagenomic or deep-sequencing raw data can provide insights into disease spread and pathogen spillover potential [100]. Modern tools based on AI/ML algorithms harness the frequency with which mutations are introduced into pathogen genomes to predict the timing of infection, identify nucleotide signatures that characterise epidemiologically linked individuals, or resolve directionality in transmission events [101]. During the COVID-19 pandemic, pathogen genome data sets grew in size substantially, from a few thousand genomes in prior epidemics to tens of millions of genomes. As a result, novel tools for the rapid analysis of large genomic data sets were built and are now available for future pandemics [102,103].

As the global climate changes, the distribution and incidence of climate-sensitive infectious diseases such as dengue and cholera is expanding [104] and the analysis of these dynamics could be enhanced by AI. Collection and incorporation of past, current, and likely future climate data into epidemiological models can improve our understanding of key epidemic dynamics and the reliability of predicted short- and long-term epidemic trajectories. Other opportunities arise when AI is used to improve short and medium-term weather forecasts; recent work has demonstrated the potential for better climate modelling using physics-based models that incorporate aspects of AI [105]. This will improve the evaluation of the impact of climate on infectious diseases, which results from changes in host and vector species distributions, or from changes in human behaviour and contact patterns.

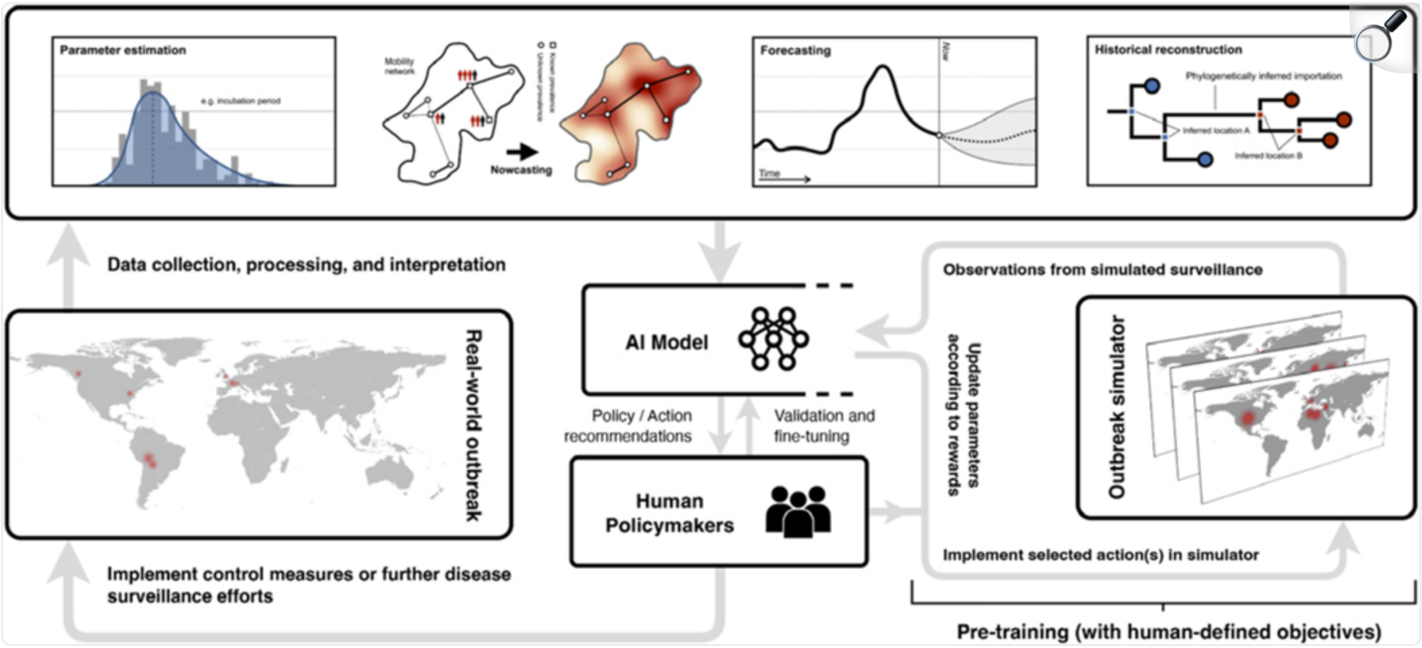
AI-driven disease modelling for policy

Public health decision making

During international health crises, such as the emergence of a novel pathogenic infectious disease, countries face complex decision-making challenges across sectors (e.g. health, economics, education, social well-being) in order to contain and mitigate disease spread. International travel and urbanisation mean that today's epidemics occur in tightly connected populations and can lead to pandemics within weeks of their emergence [106]. Decision makers must assess the cost and benefits of policies that are implemented to protect public health by maximising their efficacy whilst minimising negative and unintended consequences, all under great uncertainty. AI approaches that focus on developing generalised systems that can provide decision support [107]. Consequently, the mechanics of how public health policy decision making is done may fundamentally change with the adoption of advanced AI.

It is common for there to be an interaction between infectious disease modellers and policymakers [31], with questions and feedback from policymakers being communicated to the modelling community, sometimes coordinated through scientific advisory committees. In these interactions, time plays a critical role: decision-makers are often presented with large amounts of data and results that may be inadequately summarised at the appropriate level and not always directly relevant to the decision at hand. Meanwhile, modellers face the challenge of developing increasingly complex models to address a wide range of questions, for which characterising a clear objective or task can be sometimes difficult (Fig. 3). Due to the exponential nature of epidemic growth and consequent risks, decisions often need to be taken urgently, based on the best possible evidence available at the time. AI is likely to be a critical tool for improving the speed of modelling for informing policy making. New ML and AI approaches can reduce greatly the runtime of epidemiological models, enabling a more thorough exploration of scenarios and their statistical uncertainties. Further, LLMs could provide bespoke summaries of complex quantitative models that are tailored to a decision maker's preferences; not everyone likes graphs [108].

Figure 3: Iterative approach to public health decision making.



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Conjectured use of AI to optimise the design and implementation of effective control measures during a hypothetical multi-country disease outbreak. Data collected from disease surveillance is processed and analysed by an AI agent, potentially integrating other AI models for parameter estimation, nowcasting, and forecasting of epidemic trajectories, as well as reconstructing historical transmission events based on current observations (top row). Prior to deployment of the AI model, it is trained using an outbreak simulator which simulates the spread of infectious diseases and the effects of different control measures; the model learns from these simulations by updating its model parameters according to the outcomes of the simulations and corresponding rewards (bottom right). Insights from the empirical data are used by the trained model to inform and recommend the most effective control measures, with validation and feedback from human policymakers and stakeholders to fine-tune alignment between model objectives and societally beneficial criteria. This is followed by the execution of public health actions (for example, whether to implement recommended or alternative control measures, or deploy further disease surveillance efforts) made ultimately by human policymakers (bottom left).

Markov Decision Processes (MDPs) and Reinforcement Learning (RL) [109] are promising theoretical frameworks in which novel data types, prediction models, and expert feedback can be integrated to enable rational and more timely decision-making during epidemics (Fig. 3). This has become possible due to advances in the speed at which models can be run and updated (see section on Modelling the mechanisms of disease spread). Even though RL frameworks are attractive for tackling multi-country disease outbreaks, they require clear definitions of objectives and rewards, including enumeration of the costs and benefits of different policy actions (RL is particularly useful in robotics and strategic multiplayer games). Incorporating public health objectives, which often change and are less likely to be quantifiable into flexible models which can incorporate human-feedback might provide attractive opportunities for future science–society-policy interactions. Close collaboration between decision makers, civil society, and the scientific community is necessary therefore to advance socio-technical decision making frameworks capable of commanding well-founded public trust and confidence for epidemiology, in advance of a future global pandemic [110].

It remains an open question whether today's RL models (or other algorithms for decision-making) will be able to reconstruct and predict outbreak trajectories with sufficient certainty to be practicable, and the answer will be contingent on the quality and timeliness of available data. Unintended and often unanticipated consequences from interventions, such as the evolution of drug resistance [111] or behavioural changes in response to perceived high infection risks, have yet to be incorporated systematically into epidemic models ahead of their deployment in decision frameworks [112]. Challenges remain in evaluating the costs of interventions, which are deployed in combinations and frequently change through time or among locations. Incorporating this heterogeneity requires better methods in AI and causal inference, including emulated trials, as well as the use of hybrid models that integrate observational and causal information with simulations, to form causal "digital twins" that permit counterfactual questions to be answered [113].

Ethics in the development of AI tools

There are strong ethical reasons to support scientific efforts to explore how AI might improve infectious disease prevention and control efforts. Such improvements could save lives and reduce the burden of infectious disease. However, the successful and appropriate deployment of AI tools will depend crucially on the integrated identification, careful analysis, and resolution of key ethical challenges. Some of these are common to the application of AI in all fields and contexts, such as concerns regarding transparency, accountability, fairness, privacy, and avoidance of deepening of existing inequities [114].

Others will be more specific to the use of AI tools for infectious disease in the context of pandemic preparedness and prevention. These include the importance of equitable and fair practices for the collection, storage and sharing of data, with a focus on protecting the interests of disadvantaged communities that have been historically impacted disproportionately by infectious diseases, such as those in low and middle-income countries. Some of these ethical concerns are not specific to AI, but related to the conduct of the infectious disease surveillance upon which it depends, and its likely impact on poor and marginalised communities [115]. An important question in this regard will be what is, or should be, owed to communities that experience public health interventions with potential benefit to public health globally, but which may cause harmful impacts locally (e.g. privacy infringements, cordons sanitaires). One key ethical question, whether AI tools are being shared equitably for use by public health authorities, is likely to raise broader questions of capacity building. Making AI meaningfully accessible to all that need it will require the development and sharing of expertise through collaboration and the access to computational resources. An important and interesting ethical question in the context of the opportunities outlined in this paper might be whether the use of AI in public health policy could potentially be used to better minimise the impacts of interventions on disadvantaged communities.

A second broad class of ethical questions concerns the ways in which AI tools are to be deployed in the design and implementation of public health policy. An important lesson from COVID-19 was that all policy decisions implicitly or explicitly embody value judgements that have a moral component, for example: Who should get vaccines first? Under what conditions and to what extent can personal liberties be justifiably impinged upon in order to reduce transmission? What level of privacy should be maintained in the use of digital contact tracing [116]? It is vital that such judgements are subject to deliberation, clearly justified, and are accountable. The inclusion of AI in such processes amplifies the need for explainability, fairness, and accountability in data-informed public health policy. AI must support rather than undermine local decision-making and autonomy in public health and no technical solution will be able to fully quantify the trade-offs between the potential harms and benefits of interventions. It is vital as a component of this, and a core requirement for well-founded trust and confidence, that the development of public policy and the uses of AI in it are informed by inclusive, meaningful public engagement, as well as inclusion of public health and clinical perspectives. There will likely be an interdependence between public health deployment and on-going research and development of AI tools for new sources of data, such as environmental / wastewater surveillance. This raises important questions about the role of research as a component of effective and responsible public health and the ethics of conducting research in global health emergencies.

Public health communication

Misinformation online (and disinformation campaigns that deceive and mislead audiences) can threaten the implementation and success of public health strategies against epidemics [117]. The WHO highlighted the importance of fighting misinformation during the COVID-19 pandemic and has outlined plans for reporting online misinformation [118,119]. AI could be used to identify and address misinformation and to empower effective public health communication [117]. Due to the variety, volume, and speed of modern online communication, spanning text, images, audio and video, it is impractical to use traditional statistical approaches to investigate the dynamics of digital data, instead, AI tools will support the analysis and synthesis of accurate and reliable information about health from the digital domain.

Generative AI models can provide real-time insights into information relevant to public health [120] and how public sentiment changes during an epidemic, potentially measuring the likelihood of adoption or adherence to public health measures and complementing more traditional survey information. Including the public's feedback into AI models through LLM-based generative agents in individual-based epidemic models has been proposed and might enable those models to learn the public's perception of, and behavioural responses to, epidemic events [121]. Researchers have used NLP to analyse data about exposure to misinformation and vaccine-sceptical messages shared on social media and to provide estimates of the impact of these messages on vaccine hesitancy [122]. Extracting the complex reasons for vaccine hesitancy from quantitative data remains challenging, however [57].

There are several risks in using AI to inform or support public health communication. Careful attention is needed to avoid or reduce bias, which is typically categorised into data-driven, algorithmic, and human biases [123]. AI algorithms are by definition dependent on large training data sets, but biomedical data sets have historically excluded certain populations, including women and minority ethnic groups [124]. Therefore, biases can lead to 'ethical mistakes or misunderstandings' [125]. There is a risk in damaging public trust in public health information, with AI chatbots for instance being shown to be less popular with the public for delivering health advice [126]. Another risk relates to the propensity for LLMs to produce false information, often referred to as 'AI hallucinations' [127] but principled statistical approaches show promise in the systematic identification of these [128].

Open data & explainability

Open biological data linked to disease outcomes have powered many recent advances in the medical sciences, and have improved disease at the individual level [129,130]. For some non-communicable diseases, creation and maintenance of open databases utilising countries' centralised digital health data has been supported by governments, long-term research funding, and industry cooperation. Rigorous safety and ethical reviews should accompany and govern the access to and use of such centrally-maintained databases. Further, it is important to adopt safeguarding processes and controls for the safe release of AI models trained on sensitive data.

In infectious disease modelling the data landscape is more fragmented and harmonisation of infectious disease surveillance data across countries remains a major challenge [131]. The World Health Organization (WHO) highlighted the promotion and growth of digital health and innovation in their recent digital health agenda [132] and with the adoption of digital health, new avenues for the use of AI are in reach. Recent studies have shown the potential for software tools to integrate modern optical character recognition (OCR), natural language processing (NLP), and LLMs in order to enhance the accuracy of and speed of data extraction from semi-/unstructured sources, such as portable document formats (PDFs), press releases and situation reports [133,134]. These approaches promise to improve standard epidemiological practices, such as the metaanalysis of data from the medical literature [135] and multi-country analyses that leverage structured databases. Furthermore, basic epidemiological research tasks such as systematic reviews can be time consuming, and LLMs provide a simple interface to support the collection and summarisation of research articles [136].

Some large foundation models have been made open-source by technology companies and there is potential for them to be rigorously audited [137] and used as pretrained models for further fine-tuning in infectious disease epidemiology. For example, fine tuning pipelines such as Retrieval-Augmented Generation (RAG) could make AI models more reliable and support the extraction of valuable insights from privately stored data [138].

Further discussion is needed to understand how international, sensitive and multimodal data might be paired with novel AI models in safe and distributed hardware architecture, and how AI computing can be made more environmentally sustainable. Surveillance systems need to be linked centrally or via decentralised computational infrastructures in a secure manner, and with privacy and governance structures that can accommodate complex data sharing arrangements. An example of this is federated learning, in which partial model training occurs across multiple local servers without sharing sensitive data externally, with the results iteratively merged on a central server [139,140]. Coordinating across multiple databases, however, can become burdensome. Further, there is a need to follow existing principles to ensure that data and digital assets are comparable and can easily be deployed across multiple settings (e.g. the FAIR principles: findability, accessibility, interoperability, and reusability [141]). Importantly, data science approaches to health data require careful ethical oversight, as recently reviewed in the context of research in Africa [142]. The successful application of AI algorithms to individual health data requires trust and a shared understanding that AI use benefits the individuals contributing their data, as well as others. The challenges of data sharing and access are not limited to health data alone but extend also to data relevant in predicting health outcomes, including data on individual behaviours (e.g. human mobility data) that are often owned by companies and in some cases of commercial value [143].

Lastly, algorithms should be explainable, meaning the predictions made by the model can be interpreted by users and stakeholders alike. Unlike mechanistic models which are explainable by construction, foundation models are not built with interpretable inductive foundations and remain “black boxes” to users [144]. Quantitative measures for explainable AI have been developed, and efforts have been made in the infectious disease literature [145] but no comprehensive guidelines for the field have yet been published. Promising areas for future exploration are causal inference methods in statistics and ML [146].

Limitations of AI in ID modelling

We note at least three fundamental limitations of AI that will limit its applications to epidemic modelling. First, current models struggle with explainability, which constrains their ability to provide mechanistic insights into the transmission process and their power to generalise beyond previously observed data and scenarios. Second, AI models that are designed to support general tasks, often with a text or voice chat interface, currently do not include specific models for answering or communicating about key epidemiological questions and concepts. Integrating single task models into more general foundation models might represent an approach to developing a future general “AI-Infectious Disease” assistant. Third, capable AI models are currently trained by large technology companies at huge costs. Waiting for their deployment to then fine-tune them to answer specific tasks is not sustainable, nor equitable.

Recommendations for AI in ID epidemiology

The risks of AI have been discussed at length [107] but their potential to transform science is unquestioned. When viewed through a broad lens, AI can be seen as a continuation of the rich legacy of mathematics, statistics and computer science in the field of infectious disease epidemiology. Advances in AI have the potential to both complement existing approaches and to inspire entirely new epidemiological methods. Examples of potential breakthroughs in AI epidemiology include pretraining large and foundation models, zero-shot and few-shot learning approaches that require minimal data, innovative new data architectures, and computational environments to run and disseminate these models efficiently. In part, these breakthroughs will be enabled by the embedding of fundamental biological knowledge and constraints within AI models [71], which echoes developments in the application of AI to other fields [61,147].

The benefits of AI for public health, however, will be critically dependent on availability, accessibility, representativeness of data and a firm ethical framework. Even though more data has become available during health emergencies, especially for COVID-19, routine surveillance data for infectious diseases often remains siloed and inaccessible to the broader community, prohibiting the use of these data for improved disease modelling. The substantial volume of existing data means, however, that advances aided by AI will be likely generated from existing openly-available data sets. More broadly, the quality standard for statistical models used in health domains should be higher than for other fields, especially when understanding of cause-and-effect is relevant to treatment and policy decisions and when problems such as poor model calibration could cause significant patient harm [148]. The development of benchmarks to evaluate novel methods is therefore critical for building trust and for highlighting where AI might provide the greatest impact in the field. In part, this is why statistical models in medicine are guided not only by predictive accuracy, but also value calibration, explainability, causality and a rigorous theoretical basis. We join the AI community in recommending that the fundamental theory underpinning current AI models is studied further [149]. We advocate for robust data transparency and ethical sharing, and a detailed study of a range of biases. Finally, we support justifiable barriers of entry to including models into clinical practice, and research into robustness, training and evaluation of popular AI models.

Mechanistic models in infectious disease epidemiology continue to be developed independently of advances in AI. In this perspective we've highlighted key areas where AI could provide either incremental or potentially transformative improvements in modelling epidemic dynamics. However, evaluation of the true value added by AI approaches is difficult, and should include often-overlooked considerations such as the environmental costs associated with training complex AI models. There have been recent examples in which simpler AI methods do not consistently outperform their mechanistic counterparts [37]. Even when AI models do show significant empirical improvements, translating these models into effective government policy requires considering many factors beyond mere data fit [31]. While the potential of AI models is evident, their ultimate value will depend on demonstrating a clear opportunity cost advantage over existing approaches—which has yet to be proven definitively.

Open access to and sharing of data, analyses and results in a collaborative and transparent environment can greatly benefit the success of public health campaigns for infectious disease control [150]. In contrast, however, the scale and cost of training foundation AI models is still increasing [151] and is prohibitive for most. Whilst we hope that novel ideas to mitigate this barrier will emerge, the current suite of AI models should be more transparent in their code, architecture and data. Likewise, the development of transparent and open-access repositories of pathogen-specific data (e.g. genomic sequences), environmental data (e.g., temperature, land cover) and behavioural data (e.g. human mobility patterns) will remain central for training models and reducing data access inequities. Finally, demonstrating the effectiveness of AI in improving policy decisions that benefit population health remains a notable challenge. For AI to be successful in that regard, coming years must see growth in close collaborations between research, policy and society [152].

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Footnotes

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